Mississippi Valley State University

CS 341 — Discrete Structures

Lecture 42: One Subject

Cumulative Synthesis

The last class meeting

Name: _____ Date: _____

What this session is

Not a list of topics — the review sheet has that. This is the argument that the fifteen weeks were one subject rather than eight. **Bring questions**; we will work whatever you ask after the first twenty minutes.

1. Four ideas that never left

The asymmetry

Proving a universal claim takes an argument about an arbitrary element. Refuting one takes a single witness. Existential claims reverse it.

Where it appeared: $n^2 - n + 41$ in Week 1; satisfiability in Week 3; $\forall$ and $\exists$ in Week 4; set identities in Week 5; relation properties in Week 6; injective and surjective in Week 7; counterexamples and existence proofs in Week 10; induction in Week 11.

Counting the same thing twice

If two expressions count one collection, they are equal.

Where it appeared: $|\mathcal{P}(A)| = 2^n$ in Week 4; Pascal's identity and the Binomial Theorem in Week 13; and the First Theorem of Graph Theory in Week 14 — count (vertex, incident edge) pairs by vertex, then by edge.

2^n

Independent binary choices multiply.

Binary passwords on Day 1. Truth-table rows in Week 1. SAT's blow-up in Week 3. Power sets in Week 4. Venn regions in Week 5. Relations on a set, 2^{n^2}, in Week 6. Graphs on n labelled vertices in Week 14. And run backwards on Wednesday, it became $\log_2 n$ — the cost of binary search.

Pick the extreme object

Take a smallest counterexample, a shortest walk, a longest path. Then show it cannot exist, or cannot be improved.

Well-ordering in Week 11. Theorem 12.26 in Week 14. Theorem 13.5 on Monday.

2. The dependency chain

Each unit was built from the one before, not placed beside it.

Result	What it was built from
Set operations (Wk 5)	$\wedge$, $\vee$, $\sim$ — the laws are Week 1's.
Set equality proofs (Wk 5)	Biconditional needs two halves (Wk 2).
$\emptyset \subseteq A$ (Wk 5)	Vacuous truth (Wk 2).
Equivalence classes partition (Wk 6)	Partitions (Wk 5).
One-to-one test (Wk 7)	The contrapositive, Theorem 1.46 (Wk 2).
Pigeonhole (Wk 12)	Contrapositive of Result (5.3) (Wk 7).
A tree has $n-1$ edges (Wk 16)	Induction (Wk 11) plus Theorem 13.5.

The one that still surprises people

In Week 3, Cantor's $A_g = \{a \in A : a \notin g(a)\}$ reduced to asking whether $a_0 \in A_g$ — contradictory either way.
That is *"this sentence is false"* from Lecture 1, which we set aside as not a statement. It returned as the proof that there is no largest cardinality, and it is the shape of the halting-problem argument you will meet in theory of computation.

3. The five proof methods, and when each applies

Method	Reach for it when...
Direct	the hypothesis unfolds into a usable equation.
Contrapositive	the hypothesis is hard to use and the conclusion is easy to negate.
Cases	the hypothesis is too weak, or is a disjunction.
Contradiction	there is no implication to reverse.
Induction	the claim is indexed by the positive integers.

These are not ranked. None is more rigorous than another. The choice is practical, and the signal is almost always in the hypothesis.

4. What to do between now and the exam

In order of value

1. **Rework the problems you got wrong**, on homework, quizzes, and the midterm. Your own errors are the highest-value material you have.
2. **Write out the definitions from memory**, then check them. Most lost points on the midterm traced to a definition stated loosely.
3. **Do the practice final under timed conditions**, once, before looking at the key.
4. Then work the review sheet's self-test.

Re-reading notes feels productive and is the least effective of these.

Questions for today

Write down what you want worked at the board. We will take these in order.

Before the exam

Homework 14 is due tonight, Friday, December 4, 11:59 p.m.

Final exam: December 7–11. Check Canvas for the time and room. Cumulative, with the emphasis described on the review sheet.

A closing note

You started this course being told that checking forty cases is not a proof. You end it able to prove statements about infinitely many integers in half a page.
That is the whole of it. Thank you for a good semester.